SOP MANUFACTURING 1010BC

[CONFIDENTIAL]
[UTTHUNGA]
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CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

SL. No.	Document Version No.	Change Description	Reason For change	Date Prepared By	Date Reviewed By	Date Approved By
1	Issue B Version 1.0	Initial Draft	New process	Vanavee 02-Jul-25	Mallikarjuna B N 07-Jul-25	Bipinlal P B 11-Jul-25

1. Purpose

The purpose of this document is to define and standardize the process flow and steps of Fusion4 1010BC product in line with the organization quality standards.

2. Scope

This procedure covers assembly, testing, inspection and delivery of Fusion4 1010BC product. In this document, the terms production and manufacturing are used synonymously.

3. Abbreviations

a. BOM – Bill Of Materials

b. EOL – End Of Line

c. FG – Finished Goods

d. OTTR – On Time To Request Ratio

4. Terms and Definitions

a. OTTR: Number of on time deliveries as per customer promise dates in evaluation period / Total number of deliveries expected in evaluation period * 100%

b. EOL: Testing carried out to check unit functionality in simulation mode

c. Andon: Visual management tool used in manufacturing to alert operators to production issues allowing for quick problem identification and resolution

5. Responsibilities

a. Production Supervisor

- Create production orders in ERP.
- Check materials status from ERP and review the shortage list.
- Raise the Andon for critical shortage parts.
- Confirm FG date.
- Forward pick list to stores for material issue production order wise 1 day in advance.
- Close production order in ERP and move to FG.

b. Production Operator

- Carry out production as per standard operating procedure.
- Fill Production Work instructions and checklist stage-wise.
- Record any manufacturing defects (workmanship, drawing, material shortages) found during assembly/testing process on Production Work instructions sheet.
- Verify material shortage and completeness of the product w.r.t. Production Work instructions and Checklist and Test Reports.
- Assemble and test the units as per test procedure.

c. Quality Engineer

- Perform incoming material quality check.
- Perform finished product quality check.

6. RACI

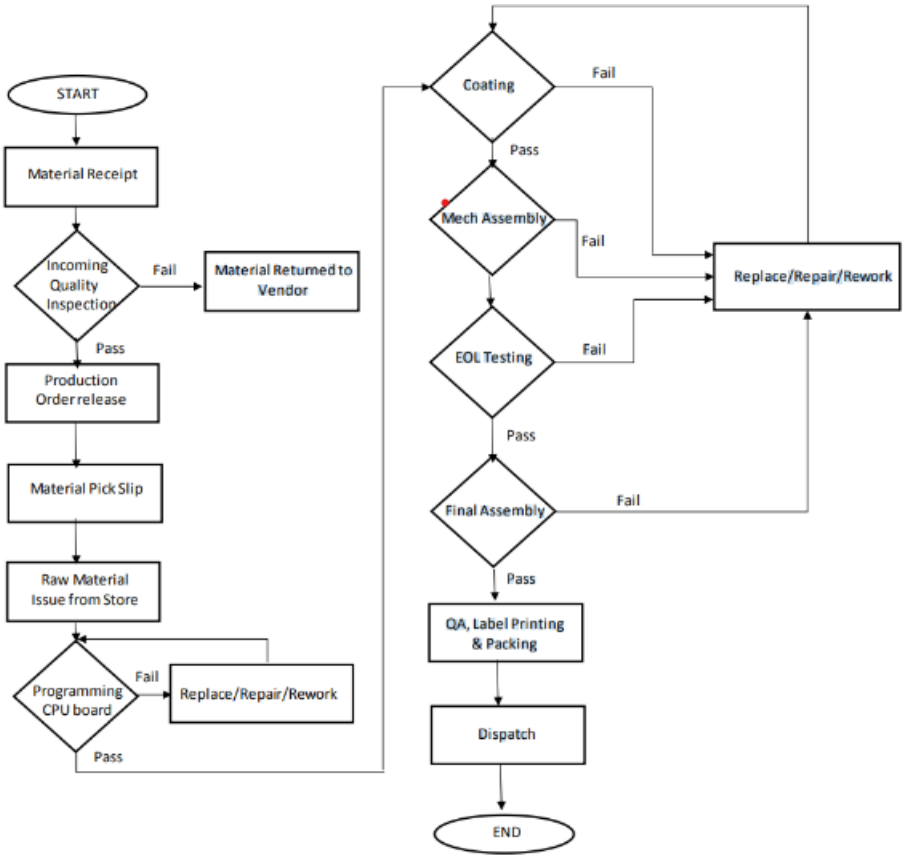
Process	Responsible	Accountable	Consulted	Informed
Production Order Creation	Production Supervisor	Production Manager		
Material Planning	Store Supervisor	SCM		
FG Date Confirmation	Production Supervisor	Production Manager		
Material Issue	Store Supervisor	Production Manager		
Production	Production Operator	Production Supervisor		
Assembly/Testing	Production Operator	Production Supervisor		
Quality Check	Quality Engineer	Production Manager/ Supervisor		Guidant Quality
Close Production Order	Production Supervisor	Production Manager		

7. SIPOC

Supplier	Input	Process and Control		Output	Customer
		Process	Control		
Vendor	Raw Materials	Material Receipt and Incoming Quality Check	Quality Inspection	Approved Materials	Production Department

Production Planning	Production Order (PO), Bill of Materials (BOM), Forecast	Production Order Creation and Material Planning	ERP System, Alert for Critical Shortages (Andon as alternative)	Production Order, Material Pick Slip	Production Supervisor
Stores	Store issue slip	Material Issue	ERP System	Issued Materials	Production Operator
Production Operator	Issued Materials, Production Work instructions, Checklist	Assembly, Testing, Calibration, EOL Testing	Standard Operating Procedures, Production Work instructions	Assembled and Tested Units	Quality Engineer
Quality Engineer	Assembled and Tested Units	Quality Check	QA Inspection Report, Functional Test report	Finished Product	Customer
Production Supervisor	Finished Product	Close Production Order	ERP System	Closed Production Order, FG (Finished Goods)	Stores

8. Overall Process Flow Chart



9. Process Description

Note: In this process description, the ERP system is referred to. However, if the ERP system is not available, there must be a central repository for tracking Production orders.

9.1 Material Receipt

a. Receive materials from suppliers.
b. Verify the quantity and quality of received materials against the purchase order in line with the procurement process
c. Record the receipt of materials in line with the inventory management process

9.2 Incoming Quality Check

a. Inspect the received materials for any defects or damages.
b. Perform quality tests as per the quality plan.
c. Accept materials that pass the quality check and move them to the storage area.
d. Return or request replacement for materials that fail the quality check.

9.3 Production Order Creation

a. Create a production order in the ERP system.
b. Ensure the production order includes all necessary details such as product specifications, quantity, and due date.

9.4 Material Planning

a. Check the material status in the ERP system.
b. Review the shortage list and raise an Andon for critical shortage parts.
c. Confirm the final goods (FG) date.

9.5 Material Issue

a. Forward the pick list to stores for material issue production order-wise, one day in advance.
b. Ensure materials are issued as per the production order requirements.

9.6 Programming CPU Board

a. Program CPU board according to the specified procedure.
b. Ensure the correct programming tool usage and follow the standard operating procedure to ensure the board is programmed correctly.

9.7 Coating

a. Apply the coating as per the specified procedure and test.
b. If the coating fails, replace, repair, or rework the material.

9.8 Mechanical Assembly

a. Assemble the units
b. Production Operator to fill the production work instructions and checklist stage-wise.
c. If the production work instructions missing during any stages of production, stop product processing and inform Production supervisor
d. Record any manufacturing defects found during the assembly process.

9.9 Testing

a. Perform Hi-Pot, Ground Bond, and other required tests.
b. Record test results in the production work instructions sheet and checklist.

9.10 EOL Testing

a. Conduct End-of-Line (EOL) testing to check unit functionality in simulation mode.

9.11 Final Assembly

a. Final assembly as per SOS.

9.12 Label Printing & Packing

a. Print the necessary labels for the product. This includes product identification labels, barcodes, and any other relevant information required for tracking and compliance
b. Carefully pack the product following the packing and shipping SOP.

9.13 Quality Check

a. Perform a final quality check on the finished product.
b. Ensure the product meets all quality standards and specifications.
c. Record the quality check results in the quality inspection report.

9.14 Product Packing and Move to FG

a. Do the product packing as per packing procedure/SOS and move to FG.

9.15 Close Production Order

a. Close the production order in the ERP system.
b. Move the finished goods to the final goods (FG) area.
c. Ensure all records and reports are updated and stored appropriately.

10. Information Security requirements

a. **Data Protection and Privacy Compliance** - All personal and sensitive information related to organization shall be securely stored and handled according to relevant data protection laws in the organizational policies
b. **Confidentiality** - Confidential information shall not be disclosed to unauthorized individuals or parties.
c. **Secure communication** - All communication channels used (e.g., emails, messaging platforms) shall be secure and encrypted wherever required.
d. **Access Control** - Restrict access to data based on the user's role and necessity.

11. References

a. SOP-Procurement (PRO) - UTSFPROSOP
b. SOP-Inventory Management

12. Metrics

a. OTTR
b. Production Cost
c. Inventory aging

13. Annexures

1. Records Management

Record Description	Custodian	Retention Period	Type	Storage Location
Production Work Instructions	Production Head	10 years for ATEX certified and 2 years for rest	Paper/Electronic	Product Shop floor file, Server
In Process Floor Failure Report	Production Head	10 years for ATEX certified and 2 years for rest	Paper/Electronic	Product Shop floor file, Server
Traceability details	Production Head	10 years for ATEX certified and 2 years for rest	Electronic	Server
Functional Test report	Production Head	10 years for ATEX certified and 2 years for rest	Electronic	Server
Quality Inspection Check list	Production Head	10 years for ATEX certified and 2 years for rest	Paper/Electronic	Product Shop floor file, Server

2. Service Level Agreement

Production Lead Time

Quantity	1-6	7-12	13-25	>25
Weeks	4	6	8-12	Contact Factory

14. Forms

a. SOS Monoblock Assembly
b. SOS Monoblock EOL Tester
c. SOS NIDO forklift for MSC Unit Lifting
d. LAD Socket Plug Assembly
e. LID back cover assembly
f. Oven Test of SSC & SVP
g. Post-hydrostatic test LID and Display Assembly
h. Pre-hydrostatic test LID and Glass Assembly
i. Stack Assembly
j. SOS-SSC Switch Board Testing/ SVP Switch Box test
k. MSC Traceability Data